



**2014
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading Time - 5 minutes
- Working Time - 2 hours
- Write using a blue or black pen
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11 - 14
- Begin each question on a new sheet of paper.

Total marks (70)

Section I

10 marks

- Attempt Questions 1-10
- Answer on the Multiple Choice answer sheet provided.
- Allow about 15 minutes for this section

Section II

60 marks

- Attempt questions 11 – 14
- Answer on the blank paper provided, unless otherwise instructed. Start a new page for each question.
- Allow about 1 hour 45 minutes for this section

Section I

Total marks (10)

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample

$2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
correct

Question 1 Which of the following is an expression for $\int \cos^2 x \sin x dx$?

(A) $2 \cos x \sin x + c$

(B) $\cos^3 x + c$

(C) $-\frac{1}{3} \cos^3 x + c$

(D) $\frac{1}{3} \cos^3 x + c$

Question 2 A particle is moving along the x -axis. Its velocity v at position x is given by $v = \sqrt{8x - x^2}$. What is the acceleration when $x = 3$?

(A) 1

(B) 2

(C) 3

(D) 4

Question 3 If $f(x) = e^{x+2}$ what is the inverse function $f^{-1}(x)$?

(A) $f^{-1}(x) = e^{x-2}$

(B) $f^{-1}(x) = e^{x+2}$

(C) $f^{-1}(x) = \log_e x - 2$

(D) $f^{-1}(x) = \log_e x + 2$

Question 4 A particle is moving in a straight line with $v^2 = 36 - 4x^2$ and undergoing simple harmonic motion. If the particle is initially at the origin, which of the following is the correct equation for its displacement in terms of t ?

(A) $x = 2 \sin(3t)$

(B) $x = 3 \sin(2t)$

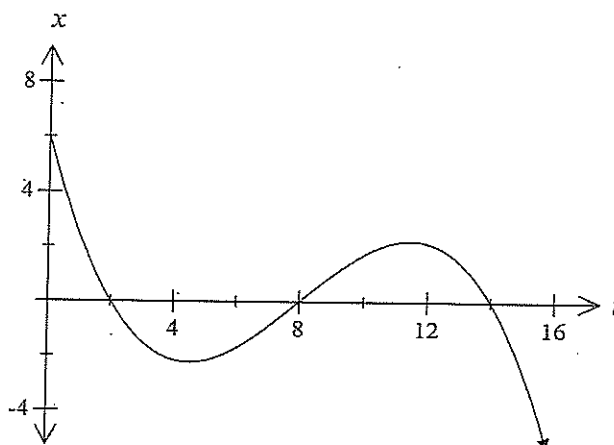
(C) $x = 2 \sin(9t)$

(D) $x = 3 \sin(4t)$

Question 5 What is the solution to the equation $\frac{\cos^3 \theta}{\sin \theta} + \sin \theta \cos \theta = 1$ in the domain $0 \leq \theta \leq 2\pi$?

- (A) $\theta = \frac{\pi}{4}, \frac{3\pi}{4}$ (B) $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$
 (C) $\theta = \frac{5\pi}{4}, \frac{7\pi}{4}$ (D) $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$

Question 6 The displacement, x metres, from the origin of a particle moving in a straight line at any time (t seconds) is shown in the graph.



When was the particle at rest?

- (A) $t = 4.5$ and $t = 11.5$ (B) $t = 0$
 (C) $t = 2, t = 8$ and $t = 14$ (D) $t = 1.5$ and $t = 8$

Question 7 The population (P) of a colony of bugs is increasing continuously at a rate proportional to the existing population. The present population is 20 000 and the population 3 months ago was 8000. If $P = Ae^{kt}$, what is the value of k ?

- (A) -0.916 (B) -0.305
 (C) 0.305 (D) 0.916

Question 8 A particle moves along a straight line about a fixed point O so that its acceleration, $a \text{ ms}^{-2}$, at time t seconds is given by $a = 4 \cos\left(2t + \frac{\pi}{6}\right)$. Initially the particle is moving to the right with a velocity of 1 ms^{-1} from a position $\frac{\sqrt{3}}{2}$ metres to the left of O . Which of the following is the correct expression for the velocity of the particle after t seconds?

- (A) $v = 2 \sin\left(2t + \frac{\pi}{6}\right)$ (B) $v = 2 \sin\left(2t + \frac{\pi}{6}\right) + 1 - \sqrt{3}$
 (C) $v = 4 \sin\left(2t + \frac{\pi}{6}\right)$ (D) $v = 4 \sin\left(2t + \frac{\pi}{6}\right) - 1$

Question 9 What is the exact value of the definite integral $\int_{\frac{2}{\sqrt{3}}}^{2\sqrt{3}} \frac{dx}{x^2 + 4}$?

- (A) $\frac{\pi}{12}$ (B) $\frac{\pi}{6}$
 (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Question 10 A particle moves in a straight line and its position at any time t is given by $x = 3 \cos 2t + 4 \sin 2t$. The motion is simple harmonic. What is the greatest speed?

- (A) 6 (B) 10
 (C) 12 (D) 20

End of Section 1

Section II

Total marks (60)

Attempt Questions 11 - 14

Allow about 1 hour 45 minutes for this section.

Answer all questions, starting each question on a new page and question number at the top of the page.

Question 11 (15 marks) Use a separate page/booklet		Marks
(a)	Find $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x}{x^2 + 4}$	1
(b)	(i) Show that $x - 2$ is a factor of $x^3 - 4x^2 + 7x - 6$	1
	(ii) Explain why $x^3 - 4x^2 + 7x - 6 = 0$ has only 1 real root.	2
(c)	Find the value of k if the roots of the equation $x^3 - 3x^2 - 6x + k = 0$ are in arithmetic progression.	3
(d)	Differentiate with respect to x : $y = \log_7 x^2$	2
(e)	Find the coordinates of the point P that divides the interval $(2, -6)$ and $(7, 9)$ internally in the ratio $2 : 3$.	2
(f)	The acceleration of a particle is given by $a = -e^{-x}$. Initially $v = \sqrt{2}, x = 0$. Find the velocity as a function of x .	2
(g)	(i) State the domain of $y = 4\cos^{-1} \frac{x}{3}$	1
	(ii) Hence, sketch the curve $y = 4\cos^{-1} \frac{x}{3}$	1

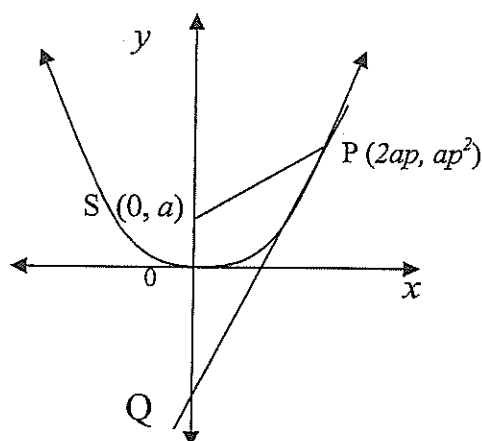
Question 12 (15 marks) Use a separate page/booklet

Marks

(a) Solve: $\frac{x}{x-2} \geq 4, \quad x \neq 2$

2

- (b) The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$ whose focus is at S. The tangent at P meets the Y-axis at Q



- (i) Derive and show that the equation of the tangent at P is $y = px - ap^2$.

2

- (ii) Find the coordinates of Q.

1

- (ii) Show that $\angle SPQ = \angle SQP$

2

- (c) A particle moves in simple harmonic motion. It starts from rest at a point 6 cm to the right of the centre of motion O. The particle has a speed of 10 cm/s, when it passes through O.

- (i) Write the expression for displacement in the form of $x = a \cos(nt + \alpha)$.

2

- (ii) Find the period of motion.

1

- (iii) Find the acceleration after 3 seconds.

1

- (d) (i) Express $\sqrt{3} \sin 2\theta - \cos 2\theta$ in the form $R \sin(2\theta - \alpha)$, where α acute.

2

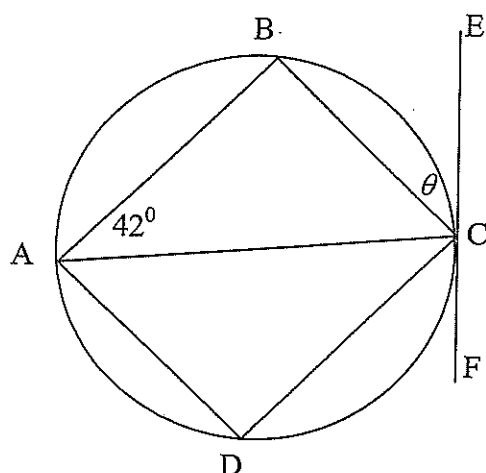
- (ii) Hence solve $\sqrt{3} \sin 2\theta - \cos 2\theta = 1$; $0 \leq \theta \leq \pi$. Answer in exact form.

2

Question 13 (15 marks) Use a separate page/booklet.

Marks

- (a) ABCD is a cyclic quadrilateral where AC bisects $\angle DAB$, $\angle BAC = 42^\circ$ and FE is a tangent to the circle at C.

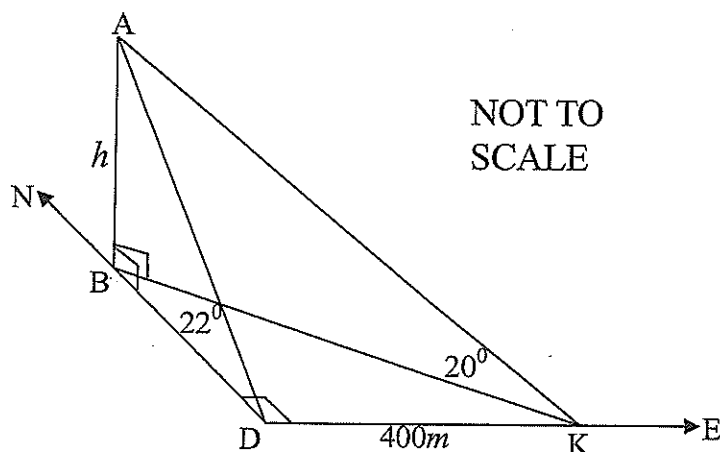


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- (i) Find the size of θ ($\angle BCE$). Give reasons. 1
- (ii) Prove that FE is parallel to DB . 3
- (b) (i) Differentiate: $y = x \sin^{-1} x + \sqrt{1-x^2}$ 2
- (ii) Hence evaluate $\int_0^{\frac{1}{2}} \sin^{-1} x \, dx$ 1
- (c) Prove that $\tan^{-1} x = \sin^{-1} \frac{x}{\sqrt{1+x^2}}$ 2

Question 13 continues on page 8

- (d) Donna is standing at D and observes the angle of elevation of the tip of a flagpole A, on top of a building to be 22° . Her friend Kate, who is standing at K, 400 metres due east of Donna, finds the angle of elevation of the tip of the flagpole to be 20° . The building is due north of Donna and B is the base of the building. The points B, D and K are all on level ground.



- (i) Show that the height (h) of the flagpole above the ground is given by:

$$h = \frac{400}{\sqrt{\cot^2 20^\circ - \cot^2 22^\circ}}$$

3

- (ii) Find the value of h , correct to 3 significant figures.

1

- (e) Using the substitution $x = u^2 - 2$ or otherwise, find $\int \frac{x}{\sqrt{x+2}} dx$

2

Question 14 (15 marks) Use a separate page/booklet.

Marks

(a) A particle is projected with speed v m/s at an angle of projection, θ to the horizontal.

(i) Derive expressions for the horizontal and vertical displacements x and y at any time t seconds after projection. Let gravity = g m/s².

2

(ii) Show that the equation of the path of the particle is given by

$$y = x \tan \theta - \frac{gx^2}{2v^2 \cos^2 \theta}$$

2

(iii) The particle has an initial speed of $2\sqrt{70}$ m/s and just clears a pole. The pole is 5m high and its base is 20m from the point of projection. Find two possible angles of projection to the nearest degree. (Take $g = 9.8$ m/s²)

2

(b) Prove by Mathematical Induction that,

3

$(n)^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all positive whole numbers n

(c) (i) Sketch the curve $y = \ln(x-2)$

1

(ii) The inner surface of a bowl is of the shape formed by rotating about the y axis, the curve $y = \ln(x-2)$ between $y = 0$ and $y = 2$

The bowl is placed with its axis vertical and water is poured in.

Show that the volume of water in the bowl when it is filled to a depth

h , where $h < 2$, is given by $\pi(4h - 4\frac{1}{2} + 4e^h + \frac{1}{2}e^{2h})$ unit³.

3

(iii) If the bowl is filled at the rate of 60 unit³ / s, find the rate at which the water level is rising when the depth of water is 1.25 units. Give your answer correct to 2 decimal places.

2



MC: C A C B B A C A A B

Q11

$$(a) \lim_{x \rightarrow \infty} \frac{\frac{3x^2}{x^2} - \frac{2x}{x}}{\frac{x^2}{x^2} - \frac{4}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{3 - \frac{2}{x}}{1 - \frac{4}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{3}{1} = 3$$

$$(b)(i) f(x) = x^3 - 4x^2 + 7x - 6$$

$$f(2) = 2^3 - 4(2)^2 + 7(2) - 6 = 0$$

$$(ii) \frac{x^2 - 2x + 3}{x-2} \cdot \frac{x^3 - 4x^2 + 7x - 6}{x^3 - 2x^2}$$

$$-2x^2 + 7x - 6$$

$$-2x^2 + 4x$$

$$3x - 6$$

$$\therefore (x-2)(x^2 - 2x + 3) = 0$$

$$x^2 - 2x + 3 \geq 0 \text{ is a factor}$$

and has no solution

$$x = \frac{2 \pm \sqrt{4 - 4 \times 1 \times 3}}{2 \times 1}$$

$$= \frac{2 \pm \sqrt{-8}}{2} \therefore \text{no solution}$$

(c) let the roots be.

$$x, x+d, x+d$$

$$x + x - d + d + d = -\frac{-3}{1}$$

$$3x = 3$$

$$x = 1$$

$$x(x-d) + x(x+d) + (x-d)(x+d) = -6$$

$$x^2 - xd + x^2 + xd + x^2 - d^2 = -6$$

$$\therefore 3x^2 - d^2 = -6$$

$$3 - d^2 = -6$$

$$d^2 = 9$$

$$x(x-d)(x-d) = -6$$

$$x(x^2 - d^2) = -6$$

$$1(1^2 - d^2) = -6$$

$$-8 = -6$$

$$\therefore 8 = 6$$

$$(d) y = \frac{\ln x^2}{\ln 7}$$

$$= \frac{2}{\ln 7} \cdot \ln x$$

$$y' = \frac{2}{\ln 7} \cdot \frac{1}{x}$$

$$= \frac{2}{x \ln 7}$$

①



$$(e) \left(\frac{2x^7 + 3x^2}{2+3}, \frac{2x^9 + 3x^6}{2+3} \right)$$

$$= (4, 0)$$

$$(f) a = -e^{-x}$$

$$\frac{d(\frac{1}{2}v^2)}{dx} = -e^{-x}$$

$$\frac{1}{2}v^2 = \int -e^{-x}$$

$$= e^{-x} + c$$

$$\frac{1}{2}(\sqrt{x})^2 = e^{-0} + c$$

$$1 = 1 + c$$

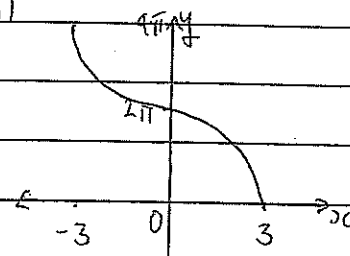
$$c = 0$$

$$\therefore \frac{1}{2}v^2 = e^{-x}$$

$$v^2 = 2e^{-x}$$

$$v = \sqrt{2}e^{-\frac{x}{2}}$$

(g)(i)



ii)

$$(1) : -3 \leq x \leq 3$$

Q12

$$(a) \frac{2x \cdot (x-2)^2}{x-2} \geq 4(x-2)^2$$

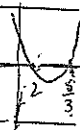
$$x(x-2) \geq 4(x-2)^2$$

$$4(x-2)^2 - x(x-2) \leq 0$$

$$(x-2)[4x-8-x] \leq 0$$

$$(x-2)(3x-8) \leq 0$$

$$\therefore 2 \leq x \leq \frac{8}{3}$$



$$(b)(i) y = x^2$$

$$\frac{dy}{dx} = \frac{4a}{4a} = \frac{2x}{2a} \therefore m = \frac{2ap}{2a} = p$$

$$y - ap^2 = p(x - 2ap)$$

$$(ii) Q \text{ at } x = 0$$

$$y - ap^2 = p(0 - 2ap)$$

$$y - ap^2 = -2ap^2$$

$$y = -ap^2$$

$$Q(0, -ap^2)$$

$$(iii) d_{sp}^2 = (2ap)^2 + (ap^2 - a)^2 = 4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2$$

$$= a^2p^4 + 2a^2p^2 + a^2$$

$$= a^2(p^4 + 2p^2 + 1)$$

$$= a^2(p^2 + 1)^2$$

②



$$d_{sp} = \sqrt{a^2(p^2+1)}$$

$$= a(p^2+1) = ap^2+a$$

$$d_{sq} = a + ap^2$$

$$= d_{sp}$$

$$\therefore \angle SPQ = \angle SQP$$

equal sides opposite equal sides

$$6n \sin \frac{\pi}{3} = 10$$

$$n = \frac{10}{6} = \frac{5}{3}$$

$$\therefore x = 6 \cos \frac{5\pi}{3}$$

$$(ii) \text{ Period} = \frac{2\pi}{n} = \frac{2\pi}{5/3}$$

$$= \frac{6\pi}{5} \text{ secs}$$

$$(C) i) x = a \cos(ut + \alpha)$$

$$\text{when } t=0, x=6$$

$$6 = a \cos \alpha$$

$$\dot{x} = -a u \sin(ut + \alpha)$$

$$\text{when } t=0, \dot{x}=0$$

$$0 = -a u \sin \alpha$$

$$\sin \alpha = 0 \therefore \alpha = 0$$

$$\therefore \cos \alpha = 1$$

$$\therefore a = 6 \quad x = 6 \cos ut$$

$$\text{when } x=0, \sin ut = 0 \therefore 6 \cos ut = 0$$

$$\cos ut = 0$$

$$ut = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$t = \frac{\pi}{2u}, \frac{3\pi}{2u}, \dots$$

$$|-6u \sin ut| = 10$$

$$6u \sin \frac{\pi}{2} = 10$$

(3)

$$ii) \sin(2\theta - \frac{\pi}{6}) = \frac{1}{2} \quad -\frac{\pi}{6} \leq 2\theta - \frac{\pi}{6} \leq \frac{11\pi}{6}$$

$$2\theta - \frac{\pi}{6} = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$2\theta = \frac{\pi}{3}, \pi$$

$$\therefore \theta = \frac{\pi}{6}, \frac{\pi}{2}$$



Q.13

$$a) i) \theta = 42^\circ \text{ (angle in alt. seg.)}$$

$$ii) \angle CAD = 42^\circ (\angle BAD \text{ is bisected})$$

$$\therefore \angle DCF = 42^\circ (\text{in alt. seg.})$$

Join DB

$$\therefore \angle DBC = 42^\circ (\text{in alt. seg.})$$

$$\therefore BD \parallel FE \text{ (alt. angles are equal)}$$

$$d) i) \frac{h}{BD} = \tan 22^\circ, \frac{h}{BK} = \tan 20^\circ$$

$$BD = \frac{h}{\tan 22^\circ} \quad BK = \frac{h}{\tan 20^\circ}$$

$$= h \cot 22^\circ \quad = h \cot 20^\circ$$

$$BD^2 + 400^2 = BK^2$$

$$h^2 \cot^2 22^\circ + 400^2 = h^2 \cot^2 20^\circ$$

$$h^2 \cot^2 20^\circ - h^2 \cot^2 22^\circ = 400^2$$

$$h^2 (\cot^2 20^\circ - \cot^2 22^\circ) = 400^2$$

$$h^2 = \frac{400^2}{\cot^2 20^\circ - \cot^2 22^\circ}$$

$$h = \frac{400}{\sqrt{\cot^2 20^\circ - \cot^2 22^\circ}}$$

$$h = 335 \text{ metres}$$

$$b) i) y = \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x + \frac{-x}{\sqrt{1-x^2}} (1-x^2)^{-1/2}$$

$$= \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x - \frac{x}{\sqrt{1-x^2}}$$

$$= \sin^{-1} x$$

$$ii) \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^1$$

$$= \left[\frac{1}{2} \cdot \sin^{-1} \frac{1}{2} + \sqrt{1-\left(\frac{1}{2}\right)^2} - (0) \right]$$

$$= \frac{\pi}{12} + \frac{\sqrt{3}}{4} - 1 = \frac{\pi}{12} + \frac{\sqrt{3}}{4} - 1$$

$$c) \text{ let } \tan^{-1} x = \alpha$$

$$\therefore x = \tan \alpha$$

$$\sin \alpha = \frac{x}{\sqrt{1+x^2}}$$

$$\sin(\tan^{-1} x) = \frac{x}{\sqrt{1+x^2}}$$

$$\therefore \tan^{-1} x = \sin^{-1} \frac{x}{\sqrt{1+x^2}}$$

ii)

$$h = 335 \text{ metres}$$

$$e) x = u^2 - 2 \Rightarrow u^2 = x+2$$

$$\frac{dx}{du} = 2u, dx = 2u du$$

$$\int \frac{x}{\sqrt{x+2}} dx = \int \frac{u^2-2}{\sqrt{u^2}} \cdot 2u du$$

$$= 2 \int u^2 - 2 du$$

$$= \frac{2u^3}{3} - 4u + C$$

$$= \frac{2(x+2)^{3/2}}{3} - 4(x+2)^{1/2} + C$$

(4)



Q14 a) i)

$$\ddot{x} = 0$$

$$\ddot{y} = -g$$

$$\dot{x} = c$$

$$\dot{y} = -gt + c$$

$$x = \sqrt{c \cos \theta}$$

$$\dot{y} = -gt + \sqrt{c \sin \theta}$$

$$x = \sqrt{c \cos \theta} + c$$

$$y = -\frac{1}{2}gt^2 + \sqrt{c \sin \theta} t + c$$

$$(7 \tan \theta - 6)(\tan \theta - 2) = 0$$

$$\tan \theta = \frac{6}{7} \quad \& \quad \tan \theta = 2$$

$$\therefore \theta = 40^\circ \quad \& \quad \theta = 63^\circ$$

when $t=0$, $x=0$, $y=0$

$$\therefore x = \sqrt{c \cos \theta} \quad \& \quad y = -\frac{1}{2}gt^2 + \sqrt{c \sin \theta} t$$

ii)

$$t = \frac{x}{\sqrt{c \cos \theta}}$$

$$y = \frac{-\frac{1}{2}gt^2}{2\sqrt{c \cos \theta}} + \frac{\sqrt{c \sin \theta}}{\sqrt{c \cos \theta}}$$

$$= \frac{-\frac{1}{2}gt^2}{2\sqrt{c \cos \theta}} + x \tan \theta$$

$$= \frac{x \tan \theta - \frac{1}{2}gt^2}{2\sqrt{c \cos \theta}}$$

$$\text{iii) } v = 2\sqrt{10}, x = 20, y = 5,$$

$$g = 9.8$$

$$\frac{20 \tan \theta}{\cos^2 \theta} - 5 = 5$$

$$20 \tan \theta - 7 \sec^2 \theta = 5$$

$$20 \tan \theta - 7(1 + \tan^2 \theta) = 5$$

$$20 \tan \theta - 7 - 7 \tan^2 \theta = 5$$

$$7 \tan^2 \theta - 20 \tan \theta + 12 = 0$$

$$\text{(b) If } n=1, (1)^5 + (1+1)^3 + (1+2)^3 = 36$$

 $\therefore$ true for $n=1$ Assume true for $n=k$

$$(k)^3 + (k+1)^3 + (k+2)^3 = 9m$$

 m is a positive integer.Prove true for $n=k+1$

$$(k+1)^3 + (k+2)^3 + (k+3)^3 = 9p$$

 p is a positive integer.

$$9m - k^3 = (k+1)^3 + (k+2)^3$$

$$\therefore \text{LHS} = 9m - k^3 + (k+3)^3$$

$$= 9m - k^3 + k^3 + 9k^2 + 27k + 27$$

$$= 9m + 9k^2 + 27k + 27$$

$$= 9(m + k^2 + 3k + 3)$$

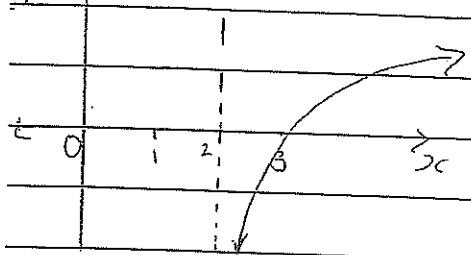
$$= 9q$$

Hence, true for $n=1$, and truetrue for $n=k$ & also true for $n=k+1$ then true for all positive integer n .

(5)

(c)

i)



$$\text{(ii) } V = \pi \int_0^2 x^2 dy$$

$$\ln(x-2) = y$$

$$x-2 = e^y$$

$$x = e^y + 2$$

$$x^2 = (e^y + 2)^2$$

$$= e^{2y} + 4e^y + 4$$

$$\therefore V = \pi \int_0^h e^{2y} + 4e^y + 4 dy$$

$$= \pi \left[\frac{1}{2}e^{2y} + 4e^y + 4y \right]_0^h$$

$$= \pi \left[\left(\frac{1}{2}e^{2h} + 4e^h + 4h \right) - \left(\frac{1}{2} + 4 \right) \right]$$

$$= \pi \left(\frac{1}{2}e^{2h} + 4e^h + 4h - 4\frac{1}{2} \right)$$

$$\text{ii) } \frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$60 = \pi (e^{2h} + 4e^h + 4) \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{60}{\pi (e^{2 \times 1.25} + 4e^{1.25} + 4)}$$

$$\approx 0.63$$

(6)